

“Explicit” time and world variables in semantics

Handout for *Time and Modality* seminar

Cian Dorr

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1 Uses of pronouns

- ‘Deictic’ uses: ‘He is a famous philosopher’.
- ‘Bound’ uses: ‘Every patient thinks that he is a famous philosopher’.
We find structural ambiguities, e.g. ‘Every miller knows at least one farmer who throws stones at his donkey’.
- Controversial anaphoric uses: ‘Every farmer who has a donkey beats it’.
- ‘Pronouns of laziness’ (?): ‘If only I had got one more question right, it would have been a car’. ‘It is ’

2 A confusion about the semantics of variables

3 The ‘intensional operator’ tradition and some of its difficulties

The need to have a specific time ‘in mind’:

- (1) I didn’t turn on the stove (Partee).

The need for ‘back-reference’:

- (2) A child was born who will become ruler of the world (Kamp)
- (3) When the war began, every soldier alive then would soon be dead (Vlach)

4 The analogy between tenses and pronouns

Deictic uses of tense: like (1).

Bound uses of tense:

- ‘When’ clauses.

Binding-type ambiguities:

- (4) Mary was in the same class as a girl whose mother was in college with someone who was working for Clinton.

Donkey-type phenomena:

- (5) Every farmer who knows that someone stole one of his chickens knows that his barn was unlocked.

5 From time to modality

- (6) Mary would have been in the same class as a girl whose mother would have been in college with someone who would have been working for Clinton.
- (7) Jane would love these flowers.

- 6 Object-language time and world variables
- 7 An argument that all propositions *expressible in natural language* are eternal and non-contingent?
- 8 An argument that all propositions are eternal and non-contingent?
- 9 Ninan's strategy: hidden initial lambda binders
- 10 An alternative strategy: unbound variables need not function like directly referential names